

## APPENDIX

*Proof of Theorem 1.* We consider standard gradient updates:

$$\begin{aligned}\mathbf{w}_{t+1}^\tau &= \mathbf{w}_t^\tau - \eta \nabla \mathcal{L}^\tau(\mathbf{w}_t^\tau), \\ \mathbf{w}_{t+1}^s &= \mathbf{w}_t^s - \eta \nabla \mathcal{L}^s(\mathbf{w}_t^s).\end{aligned}\quad (18)$$

Then, the difference between the two trajectories at step  $t+1$  can be calculated as:

$$\begin{aligned}\mathbf{w}_{t+1}^s - \mathbf{w}_{t+1}^\tau &= (\mathbf{w}_t^s - \eta \nabla \mathcal{L}^s(\mathbf{w}_t^s)) - (\mathbf{w}_t^\tau - \eta \nabla \mathcal{L}^\tau(\mathbf{w}_t^\tau)) \\ &= (\mathbf{w}_t^s - \mathbf{w}_t^\tau) - \eta (\nabla \mathcal{L}^s(\mathbf{w}_t^s) - \nabla \mathcal{L}^\tau(\mathbf{w}_t^\tau)).\end{aligned}\quad (19)$$

Taking the norm, we have:

$$\|\mathbf{w}_{t+1}^s - \mathbf{w}_{t+1}^\tau\| \leq \|\mathbf{w}_t^s - \mathbf{w}_t^\tau\| + \eta \|\nabla \mathcal{L}^s(\mathbf{w}_t^s) - \nabla \mathcal{L}^\tau(\mathbf{w}_t^\tau)\|. \quad (20)$$

Finally, we apply Eq. (20) recursively from  $t=0$  to  $t=T-1$ , noting that  $\mathbf{w}_0^s = \mathbf{w}_0^\tau$ , we obtain:

$$\|\mathbf{w}_T^s - \mathbf{w}_T^\tau\| \leq \eta \sum_{k=0}^{T-1} \|\nabla \mathcal{L}^s(\mathbf{w}_k^s) - \nabla \mathcal{L}^\tau(\mathbf{w}_k^\tau)\|. \quad (21)$$

□

*Proof of Theorem 2.* As the gradient matching objective is commonly calculated for each class separately, we define the class-wise loss for  $\mathcal{T}$  and  $\mathcal{S}$  as:

$$\begin{aligned}\mathcal{L}_{\text{GM}} &= \sum_{i=0}^{n-1} \left\| \frac{1}{|n_i|} \nabla \mathcal{L}_i^\tau(\mathbf{W}) - \frac{1}{|n'_i|} \nabla \mathcal{L}_i^s(\mathbf{W}) \right\|^2 \\ &= \sum_{i=0}^{C-1} \left\| \frac{1}{|n_i|} (\Phi_i^\top \Phi_i \mathbf{W} - \Phi_i^\top Y_i) - \frac{1}{|n'_i|} (\Phi_i'^\top \Phi_i' \mathbf{W} - \Phi_i'^\top Y_i') \right\|^2 \\ &= \sum_{i=0}^{C-1} \left\| \left( \frac{1}{|n_i|} \Phi_i^\top \Phi_i - \frac{1}{|n'_i|} \Phi_i'^\top \Phi_i' \right) \mathbf{W} - \left( \frac{1}{|n_i|} \Phi_i^\top Y_i - \frac{1}{|n'_i|} \Phi_i'^\top Y_i' \right) \right\|^2 \\ &\leq \sum_{i=0}^{C-1} \left\| \frac{1}{|n_i|} \Phi_i^\top Y_i - \frac{1}{|n'_i|} \Phi_i'^\top Y_i' \right\|^2 + \sum_{i=0}^{C-1} \left\| \frac{1}{|n_i|} \Phi_i^\top \Phi_i - \frac{1}{|n'_i|} \Phi_i'^\top \Phi_i' \right\|^2 \|\mathbf{W}\|^2 \\ &= \|\mathbf{P}\Phi(X) - \mathbf{P}'\Phi(X')\|^2 + \sum_{i=0}^{C-1} \left\| \frac{1}{|n_i|} \Phi_i^\top \Phi_i - \frac{1}{|n'_i|} \Phi_i'^\top \Phi_i' \right\|^2 \|\mathbf{W}\|^2.\end{aligned}\quad (22)$$

This decomposition reveals that distribution matching, when combined with second-order embedding alignment, provides an upper bound for class-wise gradient matching. □

*Proof of Theorem 3.* Recall that MCKP is as follows: Given  $m$  mutually disjointed groups, where group  $G_i$  contains a set of  $K_i$  items  $\{x_{i,1}, x_{i,2}, \dots, x_{i,K_i}\}$ . Each item  $x_{i,k} \in G_i$  has a value  $v_{i,k} \in \mathbb{R}_{\geq 0}$  and a weight  $w_{i,k} \in \mathbb{Z}_{>0}$ . The capacity

of the knapsack is  $W$ . MCKP aims to select exactly one item from each group such that the total weight  $\sum_i w_{i,\sigma(i)} \leq W$  is within the budget and the total value  $\sum_i v_{i,\sigma(i)}$  is maximized. Here, the mapping  $\sigma : \{1, 2, \dots, m\} \rightarrow \mathbb{Z}_{>0}$  specifies the index of the selected item in each group, i.e.,  $x_{i,\sigma(i)} \in G_i$  denotes the item chosen for group  $i$ . Based on this, we construct an instance of class-adaptive cluster allocation as: there are  $m$  classes of samples which corresponds to the number of groups in MCKP. The candidate set of cluster numbers for class  $i$  is denoted as  $\{n'_{i,1}, n'_{i,2}, \dots, n'_{i,K_i}\}$ , where  $n'_{i,k} = k$  and  $K_i = \max\{N' - (C-1), n_i\}$ . This defines the discrete search space for each class  $i$ , which is structurally equivalent to the item set  $G_i$  in MCKP. Each element in this set corresponds to an item in MCKP, where the sample count  $n'_{i,k}$  also plays a role of the item's weight  $w_{i,k}$ . Assume that we enumerate all candidate  $n'_{i,k}$ , and compute corresponding optimal clustering loss  $\mathcal{L}(\{S_j^{i,k}\}^*)$ , which corresponds to the negative value  $-v_{i,k}$  in MCKP. The total cluster budget is set to  $W$ , satisfying that  $\sum_i w_{i,\sigma(i)} = W$ . From this instance, we can derive a perfect matching for the MCKP problem. This completes the NP-hardness proof. □

*Proof of Theorem 4.* In Algorithm 1, the initialization stage incurs its main cost in the ClassWiseClustering (Algorithm 2), invoked at line 6 of Algorithm 1, where each class  $i$  performs RunKMeans on  $X_i \in \mathbb{R}^{n_i \times F}$  for  $I$  iterations. This step requires  $\mathcal{O}(n_i n'_i I F)$  time, where  $n'_i = \lfloor n_i \cdot r \rfloor$ . Hence, the overall initialization complexity is  $\mathcal{O}(I F \sum_{i=0}^{C-1} n_i \lfloor n_i r \rfloor)$ . After initialization, the algorithm iteratively alternates between SoftAllocate (Algorithm 3, invoked at line 11 of Algorithm 1), which runs in constant time, and ClassWiseClustering (line 12), which dominates the iteration cost. In the worst case, clustering must be recomputed for all classes in each iteration. Let  $\bar{n}' = \frac{1}{C} \sum_{i=0}^{C-1} n'_i = \frac{N'}{C}$  denote the average number of clusters per class. Then, over  $T$  iterations, the total cost of the search phase is  $\mathcal{O}(T \sum_{i=0}^{C-1} n_i \bar{n}' I F) = \mathcal{O}(T \cdot \frac{N N' I F}{C})$ . After convergence, Algorithm 1 performs RunKMeans( $X_i, n'_i$ ) for each class (line 22) to obtain the final partitions  $\{\{S_j^i\}^*\}$ . This post-processing cost is negligible when  $T$  is large. The time complexity of the search over  $T$  steps is  $\mathcal{O}(T \cdot \frac{N N' I F}{C})$ . Overall, the time complexity of HFILS is  $\mathcal{O}(I F \sum_{i=0}^{C-1} n_i \cdot \lfloor n_i \cdot r \rfloor + T \cdot \frac{N N' I F}{C}) = \mathcal{O}(T \cdot \frac{N N' I F}{C})$ . □